

PRACTICAL 6

Categorical variable encoding.

Consider categorical variable:

EDUCATION

Bachelor

High School

Master

1. FACTOR ENCODING

EDUCATION

EDUCATION_FACTOR

Bachelor	→	0
High School	→	1
Master	→	2

2. DUMMY ENCODING / ONE-HOT ENCODING

EDUCATION

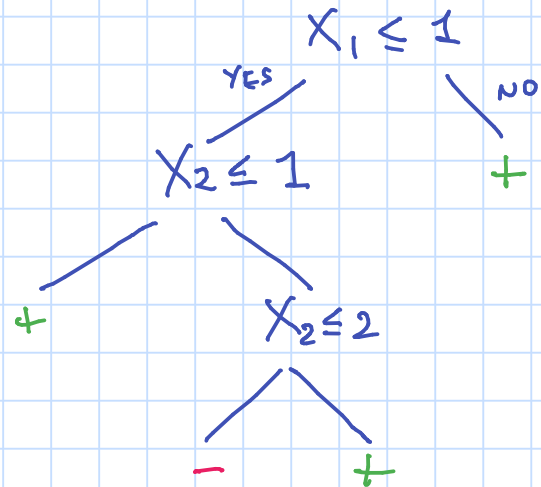
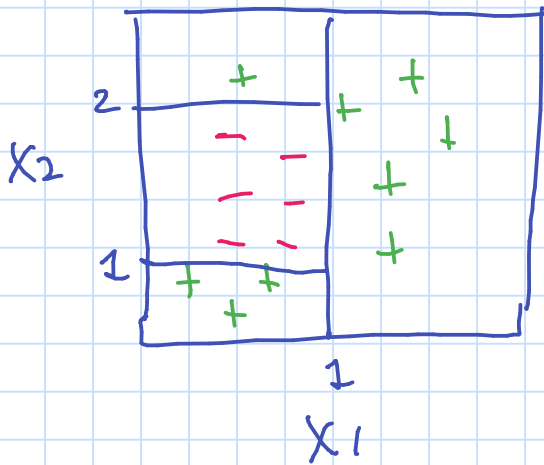
EDU_HS

EDU_BA

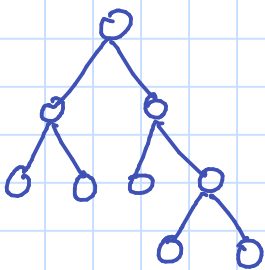
EDU_MS

Bachelor	→	0	1	0
High School	→	1	0	0
Master	→	0	0	1

Trees

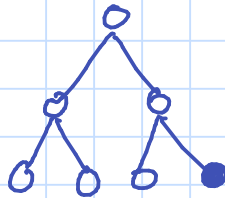


Tree pruning



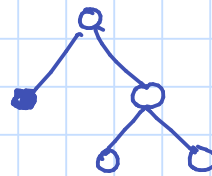
T_5

$$\alpha = 0$$



T_4

$$\alpha = 1.2$$



T_3

$$\alpha = 2.3$$



T_2

$$\alpha = 10$$



T_1

$$\alpha = 25$$

NOTE

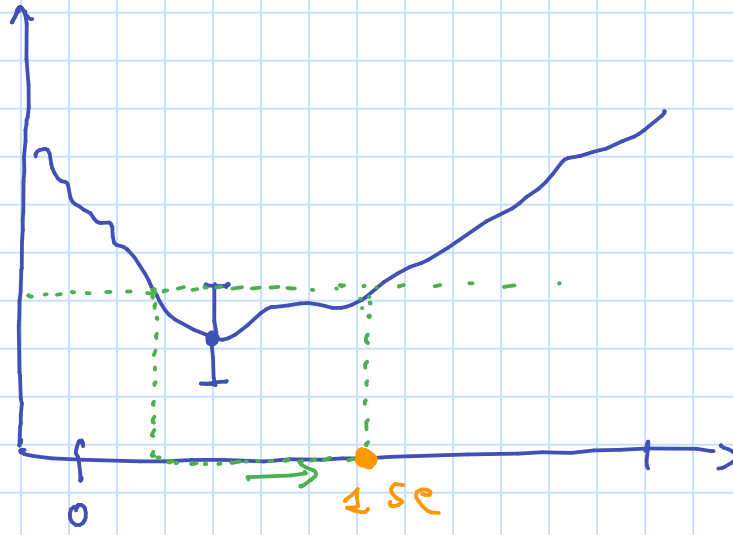
IN CLASS I WRONGLY SAID
IT'S IMPURITY MEASURE.
ACTUALLY IT'S MISCL. RATE

← Computed on training

$$R_\alpha(T) = \text{Miscl. Rate}(T) + \alpha \cdot |T|, \quad \alpha \geq 0,$$

$|T| = \# \text{ leaves}$

Misclass Error



α
↑
complexity parameter
for trees

Logistic Regression

$(x_1, y_1), \dots, (x_n, y_n) \sim$ observed data

$(x_1, y_1), \dots, (x_n, y_n) \sim$ random dataset

$$y_i, y_i \in \{0, 1\}, \quad x_i, x_i \in \mathbb{R}^p, \quad i = 1, \dots, n$$

We assume that the probability that the observation i belongs to class 1, given that it takes predictor values $X_i = x_i$ is equal to the following function

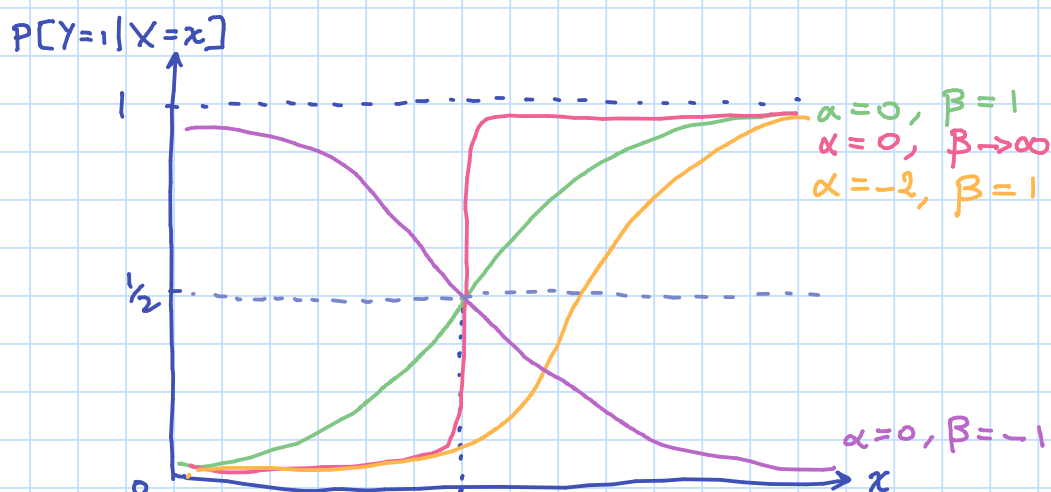
$$P(Y_i = 1 \mid X_i = x_i) = p(x_i) = \sigma(\beta^T x_i) = \frac{1}{1 + e^{-\beta^T x_i}}$$

probability $Y_i=1$ depends only on the predictor value x_i

probability changes as you change β coeff.

Ex: Suppose you have intercept and only one predictor. So,

$$P[Y=1 \mid X=x] = \frac{1}{1 + e^{-(\alpha + \beta x)}}$$



For any observation (x_i, y_i) , we can write conditional probability as

$$P[Y_i = y_i | X_i = x_i] = p(x_i)^{y_i} [1 - p(x_i)]^{(1-y_i)}, \quad y_i = 0, 1, \quad x_i \in \mathbb{R}^p.$$

where $p(x_i) = \frac{1}{1 + e^{-\beta^T x_i}}$, as defined before.

If we plug $y_i = 1$,

$$\begin{aligned} P[Y_i = 1 | X_i = x_i] &= p(x_i)^1 [1 - p(x_i)]^{(1-1)} = p(x_i) \\ &= \frac{1}{1 + e^{-\beta^T x_i}}. \end{aligned}$$

If we plug $y_i = 0$,

$$\begin{aligned} P[Y_i = 0 | X_i = x_i] &= p(x_i)^0 [1 - p(x_i)]^{1-0} = 1 - p(x_i) \\ &= 1 - \frac{1}{1 + e^{-\beta^T x_i}} = \frac{1 + e^{-\beta^T x_i} - 1}{1 + e^{-\beta^T x_i}} = \frac{e^{-\beta^T x_i}}{1 + e^{-\beta^T x_i}}. \end{aligned}$$

We can define the likelihood for all observations (x_i, y_i) , $i = 1, \dots, n$, as

$$\begin{aligned} L(\beta) &= \prod_{i=1}^n P[Y_i = y_i | X_i = x_i] \\ &= \prod_{i=1}^n p(x_i)^{y_i} [1 - p(x_i)]^{1-y_i} \end{aligned}$$


probabilities depend on β vector.

The idea is to find the optimal coefficient vector β by maximizing $L(\beta)$. However, $L(\beta)$ is hard to maximize. Recall that by taking any increasing function of $L(\beta)$ the optimal value β^* doesn't change.

So, we take $\log(L(\beta))$, that "transforms" products into sums and make the optimization easier.

We get

$$\begin{aligned} \ell(\beta) &= \log(L(\beta)) = \log \prod_{i=1}^n p(x_i)^{y_i} [1-p(x_i)]^{1-y_i} \\ &= \sum_{i=1}^n \log p(x_i)^{y_i} [1-p(x_i)]^{1-y_i} && \log(ab) = \log a + \log b \\ &= \sum_{i=1}^n \log p(x_i)^{y_i} + \log [1-p(x_i)]^{1-y_i} && \log(ab) = \log a + \log b \\ &= \sum_{i=1}^n y_i \log p(x_i) + (1-y_i) \log [1-p(x_i)] \end{aligned}$$

Annotations:

- y_i : true labels (purple arrow)
- $p(x_i)$: predicted probabilities (orange arrow)
- $\log a^b = b \log a$ (green arrow)
- depend on β (green arrow)

If I want to maximize $\ell(\beta)$, ideally, I would choose

$$\begin{cases} p(x_i) = 1, & \text{if } y_i = 1, \\ p(x_i) = 0, & \text{if } y_i = 0. \end{cases} \quad i=1, \dots, n. \quad (\text{convention: } -\infty \cdot 0 = 0).$$

However, we cannot tune the probabilities $p(x_i)$ independently for each observation. All we can do is to tune a single vector β that changes all $p(x_i)$, $i=1, \dots, n$.

Notice that:

$$1 - p(x_i) = \frac{e^{-\beta^T x_i}}{1 + e^{-\beta^T x_i}} = \frac{1}{\frac{1 + e^{-\beta^T x_i}}{e^{-\beta^T x_i}}} = \frac{1}{e^{\beta^T x_i} + 1}$$

$$\frac{p(x_i)}{1 - p(x_i)} = \frac{1}{1 + e^{-\beta^T x_i}} \cdot \frac{1 + e^{-\beta^T x_i}}{e^{-\beta^T x_i}} = e^{\beta^T x_i}$$

If we continue solving $\ell(\beta)$, we get

$$\ell(\beta) = \sum_{i=1}^n y_i \log p(x_i) + (1 - y_i) \log [1 - p(x_i)]$$

$$= \sum_{i=1}^n \log [1 - p(x_i)] + y_i \{ \log p(x_i) - \log [1 - p(x_i)] \}$$

$$= \sum_{i=1}^n \log [1 - p(x_i)] + y_i \log \frac{p(x_i)}{1 - p(x_i)}$$

$$= \sum_{i=1}^n \log \frac{1}{1 + e^{\beta^T x_i}} + y_i \log e^{\beta^T x_i}$$

$$= \sum_{i=1}^n -\log (1 + e^{\beta^T x_i}) + y_i (\beta^T x_i)$$

$$= \sum_{i=1}^n y_i (\beta^T x_i) - \log (1 + e^{\beta^T x_i}).$$

$$\begin{aligned} \bullet \log\left(\frac{1}{b}\right) &= \log 1 - \log b \\ &= 0 - \log b \end{aligned}$$

$$\bullet \log e^x = x$$